

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: **6th**

Name of the Course: **Machine Learning**

Course Code: **TCS 662**

Time : **90 Minutes**

Maximum Marks : **50**

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1																																
(a)	Explain bias, variance, and bias-variance trade-off in machine learning with the help of examples.	CO2																																
	OR																																	
(b)	Compare how logistic regression is different from linear regression. Explain regularization with the help of an example.																																	
Q2	(10 Marks)	CO2																																
(a)	Explain three different validation techniques namely Hold-out, K-Fold cross validation and Leave-one-out. Discuss their pros and cons in detail.																																	
	OR																																	
(b)	What is regularization, and why is it used in machine learning? How does regularization help in preventing overfitting?																																	
	(10 Marks)	CO2 CO3																																
Q3																																		
(a)	Consider the following training set having three different attributes namely x1, x2, x3: <table><tr><th>X1</th><th>X2</th><th>X3</th><th>Class</th></tr><tr><td>-1</td><td>1</td><td>1</td><td>Negative</td></tr><tr><td>5</td><td>1</td><td>0</td><td>Positive</td></tr><tr><td>0</td><td>2</td><td>1</td><td>Negative</td></tr><tr><td>-4</td><td>-1</td><td>0</td><td>Negative</td></tr><tr><td>0</td><td>0</td><td>1</td><td>Positive</td></tr><tr><td>9</td><td>2</td><td>1</td><td>Positive</td></tr><tr><td>2</td><td>2</td><td>0</td><td>Negative</td></tr></table> Using the K-nearest algorithm (KNN) algorithm with K=3, Classify the following points i) (1,1,2) ii) (2,1,1) iii) (0,0,0)	X1	X2	X3	Class	-1	1	1	Negative	5	1	0	Positive	0	2	1	Negative	-4	-1	0	Negative	0	0	1	Positive	9	2	1	Positive	2	2	0	Negative	
X1	X2	X3	Class																															
-1	1	1	Negative																															
5	1	0	Positive																															
0	2	1	Negative																															
-4	-1	0	Negative																															
0	0	1	Positive																															
9	2	1	Positive																															
2	2	0	Negative																															
	OR																																	

	A dataset contains 14 instances , classified as follows: I. 8 instances belong to class "1" II. 7 instances belong to class "0"											
(b)	A feature Z splits the dataset into three subsets: I. Subset 1: (3 "1", 2 "0") II. Subset 2: (3 "1", 3 "0") III. Subset 3: (2 "1", 2 "0") Calculate the entropy of the split dataset and determine the information gain.											
Q4	(10 Marks)	CO1										
(a)	What is Locally Weighted Regression, and how does it differ from the K-Nearest Neighbour approach?	CO3										
	OR											
(b)	Discuss the working of Support Vector Machine. A linear SVM separates two classes using the hyperplane: $3x+4y-10=0$ The closest support vectors to this hyperplane are at points (2,1) and (3,2) . Calculate the margin of this SVM.											
Q5	(10 Marks)	CO1										
(a)	What is Boosting in Ensemble Learning? Explain the working of the decision trees algorithm as an example of Boosting.											
	OR											
(b)	A linear regression model is given by: $\text{Prediction}(Y) = 5X+2$ Compute predicted values using the given regression model and also calculate the MSE(error) for the following dataset. <table><tr><th>X</th><th>Y</th></tr><tr><td>2</td><td>12</td></tr><tr><td>4</td><td>22</td></tr><tr><td>6</td><td>32</td></tr><tr><td>8</td><td>42</td></tr></table>	X	Y	2	12	4	22	6	32	8	42	CO2
X	Y											
2	12											
4	22											
6	32											
8	42											